Transient Freezing Governs Flat-Minimum Selection in Stochastic Gradient Descent

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Stochastic gradient descent (SGD) often converges to flatter minima with better generalization, but the dynamical event that fixes the final basin remains unclear. Existing theories mainly describe a late-stage or fixed-landscape preference induced by anisotropic noise. Here we show that basin selection is controlled by an earlier nonequilibrium process. In neural-network experiments, SGD first remains mobile among distinct solution valleys and then loses this mobility at a well-defined freezing time. Continuation-training measurements identify the valley occupied by the trajectory at different checkpoints and show that the final basin is selected when inter-valley motion arrests. Increasing SGD noise delays this freezing time and thereby increases the probability of ending in a flatter, better-generalizing valley. A minimal two-valley model with landscape-dependent noise reproduces the observed hopping, delayed freezing, and final selection. The model also resolves an apparent paradox: stronger noise can weaken the instantaneous fixed-slice preference for the flat valley while enhancing the final flat-minimum probability by shifting the freezing point. These results recast the implicit bias of SGD as a history-dependent nonequilibrium selection mechanism controlled by the duration of transient mobility.

I. INTRODUCTION

Despite its empirical success, the physical mechanisms by which deep learning selects a final solution remain poorly understood [1, 2]. Training a neural network can be viewed as optimization on a high-dimensional loss landscape $\mathcal{L}(\boldsymbol{\theta})$ driven by stochastic gradient descent (SGD), where $\boldsymbol{\theta}$ denotes the trainable parameters [3, 4]. A prevailing hypothesis is that the geometry of this landscape is closely linked to generalization: solutions lying in flat or wide minima tend to generalize better than those in sharp or narrow minima [5–10]. The central question is therefore not only why flat minima are favored, but also when and how this preference becomes a committed basin choice during optimization.

The relevant landscape is not a single basin, but a collection of competing solution valleys. Consistent with previous studies [6, 8, 11–15], we find that independent SGD runs initialized from the same point converge to multiple distinct low-loss endpoints (Fig. S1 in the Supplemental Material). Although all these endpoints achieve zero training error, they form separated clusters and are divided by loss barriers along interpolation paths. Hessian spectra further reveal flat directions along the valley floor and steep transverse directions, supporting their interpretation as distinct solution valleys. Basin selection during training should therefore be understood as a dynamical process in which the optimizer navigates among competing valleys before committing to one of them.

The stochasticity driving this navigation has a distinctive structure. Due to mini-batch sampling, the noise covariance in SGD aligns with the local Hessian, making the noise both landscape-dependent and anisotropic [16–19]. This anisotropic stochasticity is believed to bias optimization trajectories toward flatter regions of the loss landscape [16, 17, 19]. Empirical studies further indicate that increasing the overall noise strength of SGD—by raising the learning rate η or reducing the batch size B—typically leads to convergence toward flatter minima with improved generalization [7, 9, 15]. However, most existing analyses of this implicit regularization focus on late-stage or fixed-landscape regimes, where gradients are already small and trajectories have entered low-loss valleys. Such analyses do not, by themselves, explain how the final basin is selected during early training, when gradients remain large, barriers are still evolving, and the trajectory can remain mobile among competing valleys.

Multiple studies have demonstrated the critical importance of the early phase of training, showing that key properties of the final network are established long before convergence [20–24]. For instance, the "lottery ticket hypothesis" suggests that trainable subnetworks emerge very early in the training process [25]. This raises a crucial question: How does SGD navigate the complex, high-dimensional loss landscape during these formative early stages to set the trajectory toward generalizable solutions? Understanding this "early transient dynamic" is essential for a complete theory of deep learning optimization.

In this paper, we systematically study how stochastic gradient descent (SGD) escapes sharp minima and preferentially converges toward flatter regions during early training when there are significant down-hill gradients. Combining empirical observations with a tractable theo-

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retical model, we characterize the transient dynamics underlying this behavior. We identify a stochastic "valley-jumping" mechanism that enables SGD to explore multiple basins of attraction in the initial phase of optimization.

Our study provides a clear physical picture of transient learning dynamics. Anisotropic noise in SGD reshapes the loss landscape, giving rise to an emergent effective potential that systematically biases the dynamics toward flatter, more generalizable minima. Crucially, this exploratory behavior is intrinsically transient: as training proceeds, the evolving landscape geometry dynamically "freezes" the trajectory into a single valley. Higher noise levels—arising from larger learning rates or smaller batch sizes—prolong this pre-freezing regime, allowing the dynamics more time to follow the effective potential and discover flatter solutions. Taken together, these results offer a unified framework connecting stochasticity, loss landscape geometry, and generalization, and reveal an explicitly transient mechanism by which SGD preferentially selects flat minima.

II. RESULTS

A. Transient valley hopping drives convergence toward flatter minima

To directly visualize how SGD transitions between valleys during training, we designed a continuation training experiment. A network was first trained with SGD to generate a reference trajectory (Fig. 1(a); reference setting $B = 50$, $\eta = 0.05$). At selected checkpoint times t_c along this trajectory, training was branched and resumed using deterministic full-batch gradient descent (GD). Because full-batch GD removes mini-batch stochasticity, each continuation rapidly descends into the local valley determined by its starting point, without further inter-valley exploration. The continuation endpoint therefore identifies the valley occupied by the SGD trajectory at time t_c (Fig. 1(a); see Supplemental Material for details).

These results provide direct evidence that valley hopping during early training is biased toward flatter and more generalizable regions. Continuations started from early checkpoints converge to solutions with relatively high test loss and low flatness. As t_c increases, later continuations progressively land in valleys with lower test loss and higher flatness [Fig. 1(b), (c)]. We quantify flatness F as the inverse geometric mean of the top N non-degenerate Hessian eigenvalues,

$$F \equiv \left(\prod_{i=1}^{N} \lambda_i(\mathbf{H}) \right)^{-\frac{1}{N}}, \tag{1}$$

where we set $N = 10$ for the 10-class MNIST task [26]. This systematic improvement indicates that the SGD trajectory does not remain trapped in its initial basin of attraction, but instead transitions across the landscape to discover flatter and more generalizable regions.

We further found that valley jumping occurs almost exclusively before a well-defined freezing time. To identify when inter-valley mobility is lost, we compared the continuation endpoints obtained from different times. Let $\hat{\boldsymbol{\theta}}_i$ denote the endpoint reached by deterministic continuation from time t_i. Before freezing, different times may lead to distinct continuation endpoints; after freezing, later continuations instead relax to mutually similar endpoints. We quantify this endpoint similarity using the Jaccard similarity between their misclassified test sets,

$$\mathrm{Sim}_J(\hat{\boldsymbol{\theta}}_i, \hat{\boldsymbol{\theta}}_j) = \frac{|\mathcal{E}(\hat{\boldsymbol{\theta}}_i) \cap \mathcal{E}(\hat{\boldsymbol{\theta}}_j)|}{|\mathcal{E}(\hat{\boldsymbol{\theta}}_i) \cup \mathcal{E}(\hat{\boldsymbol{\theta}}_j)|}, \tag{2}$$

where $\mathcal{E}(\hat{\boldsymbol{\theta}})$ is the set of test samples misclassified by the continuation endpoint $\hat{\boldsymbol{\theta}}$. Unlike similarity measures defined directly in weight space, which suffer from permutation symmetry of neurons, this error-based similarity is invariant to such symmetries and thus provides a more faithful measure of solution similarity.

The resulting similarity matrix [Fig. 1(d)] shows a clear transition: continuation endpoints from early times can differ substantially, whereas endpoints from later times become highly similar, forming the red block in Fig. 1(d) (see Supplemental Material for details). We define the freezing time as the earliest time after which all subsequent continuation endpoints remain mutually similar,

$$t_f \equiv \min \left\{ t_c \;\middle|\; \mathrm{Sim}_J(\hat{\boldsymbol{\theta}}_i, \hat{\boldsymbol{\theta}}_j) \geq S_{\mathrm{th}}, \;\; \forall\, t_i, t_j \geq t_c \right\}, \tag{3}$$

where S_{th} is a high-similarity threshold.

Before the freezing time t_f, the continuation endpoints become progressively flatter and achieve lower test loss. After t_f, all later continuations relax to highly similar endpoints, and both flatness and test loss largely saturate [Fig. 1(e)]. Thus, the key transition is not final convergence itself, but the earlier point at which SGD loses the ability to move between competing valleys. Together, these results show that transient valley hopping drives SGD toward flatter minima before the trajectory becomes frozen.

B. SGD noise delays freezing and broadens accessible solutions

The continuation-defined freezing time identifies when a trajectory becomes committed to an endpoint class. We next ask how this commitment time depends on SGD noise, and whether delaying commitment changes the ensemble of final solutions that training can reach. This comparison is not meant to prove that t_f is the only variable controlling generalization. Instead, it establishes the empirical connection between stochastic mobility, basin commitment, and final-solution statistics that the theory below will explain.

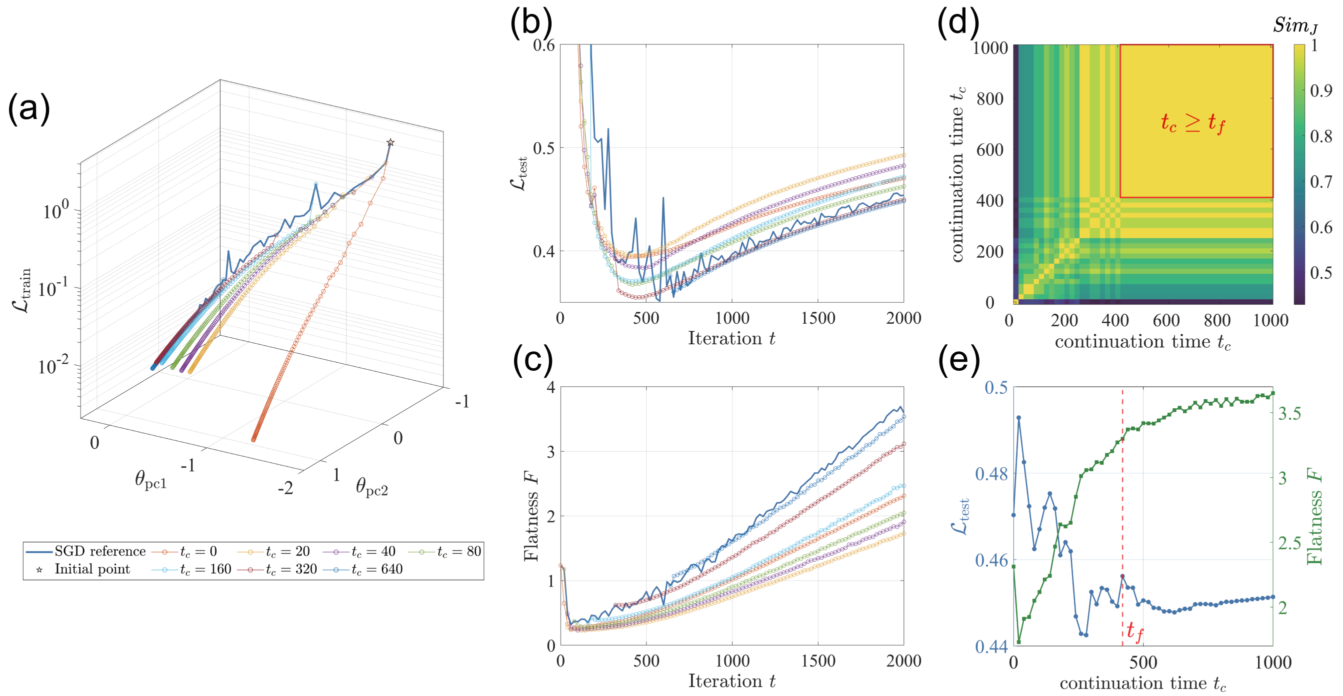

FIG. 1. Continuation training reveals transient valley hopping and freezing during SGD. (a) A PCA projection of the reference SGD trajectory (blue; $B = 50$, $\eta = 0.05$) and deterministic full-batch GD continuations branched from different checkpoint times t_c (colored curves). The vertical coordinate indicates training loss. (b, c) Evolution of the test loss $\mathcal{L}_{\text{test}}$ (b) and flatness F (c) for the reference SGD trajectory and the GD-continued trajectories. (d) Jaccard similarity Sim_J among solutions obtained from different continuation times t_c, with the red square marking the region where $t_c \geq t_{\text{freeze}}$. (e) Endpoint test loss and flatness ((evaluated at $t = 2000$)) as functions of continuation time t_c. The vertical dashed line marks the same freezing time t_f inferred from panel (d).

All measurements in this section use the same controlled training setup. We trained a fully connected ReLU network on a fixed 1,000-sample subset of MNIST, with 100 images per class, and evaluated on a fixed 200-sample test set. The architecture was $784 \to 50 \to 50 \to 10$, and the parameters were optimized by standard SGD without momentum or weight decay to minimize the cross-entropy loss. To isolate the effect of mini-batch stochasticity, all runs started from the same Kaiming-initialized weights with zero biases. We scanned learning rates $\eta \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1\}$ and batch sizes $B \in \{10, 20, 50, 100, 200, 500, 1000\}$, where $B = 1000$ corresponds to full-batch GD, and performed 20 independent repeats for each condition. The total number of iterations was scaled as $T = 100/\eta$, with trajectories recorded at 101 evenly spaced checkpoints.

Within this hyperparameter grid, increasing the learning rate η or decreasing the batch size B increases the effective stochasticity of SGD, with nominal noise scale proportional to η/B. We measured t_f with a matched-continuation protocol: each hyperparameter condition was continued using its own training learning rate, so the readout tests commitment under the dynamics appropriate to that condition. With this definition, the deterministic large-batch baseline has little mini-batch-induced redirection, and the main comparison focuses on stochastic settings with $B < 1000$. Across these settings, the rescaled freezing coordinate ηt_f increases systematically toward larger η/B [Fig. 2(a)]. Thus, stronger SGD noise keeps the trajectory mobile among competing endpoint classes for a longer portion of training before the continuation endpoint becomes stable.

Connecting this commitment time to final solution quality requires some care. With cross-entropy loss, training can continue to change margins, local sharpness, and test loss after classification accuracy has saturated and after the endpoint class has already been selected. We therefore do not treat the final network as an instantaneous readout of the freezing event. Instead, we use converged endpoints to ask which region of solution space becomes accessible when the mobile phase lasts longer. For each hyperparameter condition, we measured the RMS radius of the converged endpoint ensemble,

$$r_\theta \equiv \left[\left\langle \left\| \hat{\boldsymbol{\theta}}^{(n)} - \langle \hat{\boldsymbol{\theta}}^{(n)} \rangle_n \right\|^2 \right\rangle_n \right]^{1/2}, \tag{4}$$

where $\hat{\boldsymbol{\theta}}^{(n)}$ is the final converged endpoint of repeat n, and $\langle \cdot \rangle_n$ denotes an average over repeats. We measured this endpoint radius together with the final flatness and the upper envelope of final test accuracy.

These final-solution observables follow the same noise ordering as the freezing coordinate. High-noise conditions have larger r_θ [Fig. 2(b)], indicating that delayed freezing is associated with a broader set of converged endpoints across independent repeats. The mean final flatness increases with the same ordering [Fig. 2(c)]. By contrast, mean test accuracy is strongly compressed on this small task, so we report the best-of-repeats test accuracy only as an upper-envelope statistic [Fig. 2(d)]. This panel should be interpreted conservatively: it shows that noisier, later-freezing conditions can access high-performing endpoints, not that noise monotonically improves typical accuracy.

These observations refine the statement that more noise produces more exploration. The relevant exploration is the transient mobility that persists before the continuation endpoint becomes stable. After this arrest, later optimization mainly refines the already selected endpoint class. We therefore use t_f as a dynamical readout of the noise-controlled exploratory window, rather than as a post hoc label assigned after convergence.

The resulting picture is that final-basin statistics are shaped by two coupled ingredients: the evolving landscape encountered during training and the time at which inter-valley mobility is lost. Increasing SGD noise does not simply perturb the final basin after it has been chosen. Rather, it delays freezing, leaving the trajectory mobile until barriers, basin volumes, and flatness contrasts have evolved further. This extended pre-freezing window gives SGD a greater chance to reach flatter endpoint classes before the basin choice is locked in. In this sense, the transient noisy phase is not wasted wandering; it determines which part of the evolving landscape can still influence the final solution ensemble. The minimal model below is designed to separate the fixed-slice valley bias from the noise-dependent stopping time at which that bias is inherited.

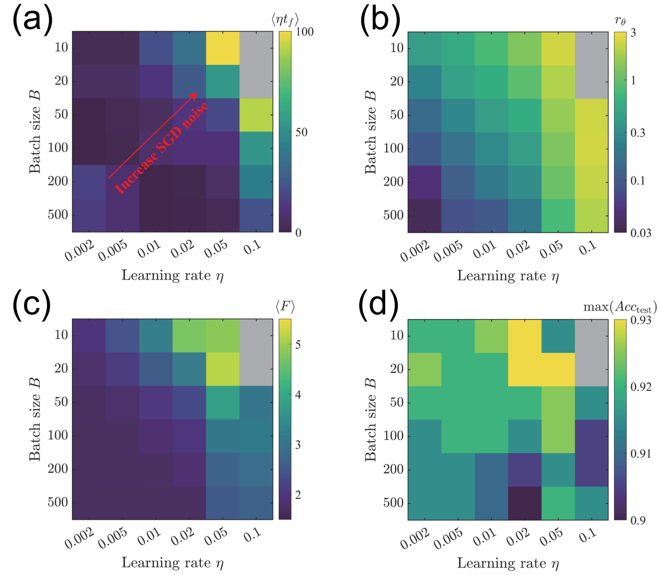

FIG. 2. SGD noise delays freezing and broadens the set of accessible final solutions. (a) Mean matched-continuation freezing coordinate ηt_f across batch size and learning rate. The axes are arranged so that the nominal SGD noise scale η/B increases toward the upper-right corner; $B = 1000$ and $\eta = 0.001$ are omitted. (b) Across-repeat RMS endpoint radius $r_\theta = [\langle \|\hat{\boldsymbol{\theta}}^{(n)} - \langle \hat{\boldsymbol{\theta}}^{(n)} \rangle_n \|^2 \rangle_n]^{1/2}$ for final-converged repeats, shown on a logarithmic color scale. (c) Mean final flatness F for the same final-converged repeats. (d) Best final test accuracy over repeats. Panel (d) is an upper-envelope statistic and is not used as evidence that noise improves typical accuracy. Gray cells indicate settings without final-converged repeats for the corresponding statistic.

C. A minimal theory of transient freezing

To isolate the mechanism, we constructed the smallest model that contains the ingredients observed in training. The coordinate y represents the downhill training direction, while the coordinate x distinguishes two competing valleys. The valley at $x > 0$ is flatter than the valley at $x < 0$, but the two valleys have equal depth relative to the separating barrier, so any selection effect is geometric rather than imposed by an energy difference. A convenient realization is

$$\mathcal{L}(x,y) = \mathcal{L}_0(y) + \begin{cases} \dfrac{x\,[x - g_1(y)]}{f_1(y)}, & x \geq 0, \\ \dfrac{x\,[x + g_2(y)]}{f_2(y)}, & x < 0, \end{cases} \tag{5}$$

where $g_{1,2}(y)$ set the valley positions and $f_{1,2}(y)$ are inverse curvatures. We denote $f_{\rm flat}(y) \equiv f_1(y)$ and $f_{\rm sharp}(y) \equiv f_2(y)$, with flatness contrast $\gamma \equiv f_{\rm flat}/f_{\rm sharp} > 1$. As y increases, the valleys separate, the barrier height $\Delta\mathcal{L}(y)$ grows, and the absolute flatness of the valleys changes. These are the empirical features needed for a freezing problem.

The dynamics follow a discrete-time Langevin update,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\,[\nabla\mathcal{L}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_t]\,, \qquad \boldsymbol{\theta} = (x,y), \tag{6}$$

with zero-mean noise whose covariance mimics the geometry-coupled structure of SGD,

$$\mathrm{Cov}(\boldsymbol{\xi}_t) = 2\sigma\,\mathbf{H}(\boldsymbol{\theta}_t). \tag{7}$$

Here $\mathbf{H}$ is the local Hessian in the stable valley region and σ sets the mini-batch noise scale. The effective noise level is $\Delta_S \sim \eta\sigma$. Thus, Δ_S controls stochastic mobility, $\Delta\mathcal{L}(y)$ controls the barrier to valley exchange, and γ controls the flatness asymmetry.

The model reproduces the empirical sequence. Early in training, barriers are small and trajectories hop repeatedly between the two valleys. As y increases, escape rates decrease and hopping becomes rare. The basin is selected when the hopping time exceeds the drift time along y. Increasing Δ_S delays this crossing, keeps the system mobile for longer, and increases the probability that the final basin is the flatter valley.

FIG. 3. Minimal two-valley model for transient freezing. (a) The coordinate y describes downhill training, while x distinguishes a flat valley from a sharp valley. The valleys separate and the barrier grows as y increases. (b) Geometry-coupled noise is stronger along high-curvature directions, mimicking the anisotropic structure of SGD noise. (c) Simulated trajectories hop between valleys early in training and freeze into one valley later. (d) Stronger effective noise Δ_S delays freezing and increases the final probability of selecting the flat valley.

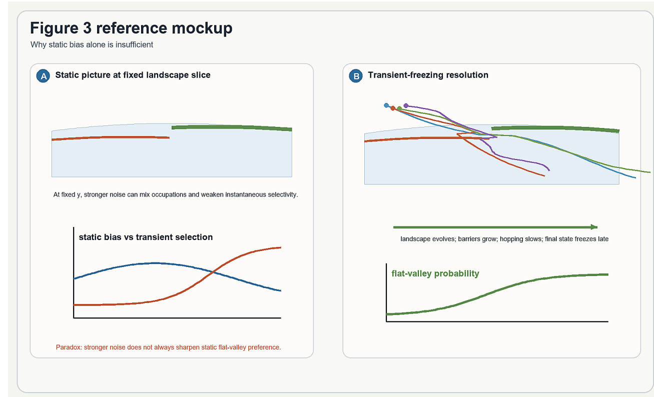

FIG. 4. Static bias and its limitation. (a) On a fixed landscape slice, anisotropic noise induces an effective loss that favors the flatter valley. (b) The corresponding quasi-steady flat-valley probability $P^{\rm ss}_{\rm flat}(y)$ is larger than the equilibrium expectation. (c) At fixed y, increasing Δ_S can reduce instantaneous selectivity by mixing the two valleys. (d) The apparent contradiction is resolved by recognizing that stronger noise delays freezing, so the final probability is inherited from a later landscape slice.

This model is not intended as a literal two-coordinate reduction of a neural network. Its role is to make the mechanism identifiable. Once competing valleys, geometry-coupled noise, and a growing barrier coexist, flat-minimum selection becomes a history-dependent consequence of nonequilibrium dynamics, rather than only a property of a fixed landscape.

D. Static bias is real but insufficient

The reduced model also explains why a fixed-landscape theory is incomplete. At an approximately fixed y, the transverse coordinate x can relax faster than the longitudinal drift. In this quasi-steady regime, anisotropic SGD noise generates a nonequilibrium preference for the flatter valley [16, 17, 19]. Let $\delta f(y) \equiv f_{\rm sharp}(y) - f_{\rm flat}(y)$. Kramers-rate estimates give the flat-valley occupation probability

$$P^{\rm ss}_{\rm flat}(y) \approx \left[1 + \gamma^{-\frac{1}{2}} \exp\left(\frac{\Delta\mathcal{L}(y)\delta f(y)}{2\Delta_S}\right)\right]^{-1}. \tag{8}$$

Because $f_{\rm flat} > f_{\rm sharp}$, $\delta f < 0$ and this expression is biased toward the broader valley. Equivalently, anisotropic noise reshapes the effective landscape by lowering the flat valley relative to the sharp one.

However, Eq. (8) also exposes the limitation of a static picture. If y is held fixed, increasing Δ_S does not necessarily make the flat valley more selectively occupied. A larger effective noise level can mix the two valleys and weaken the instantaneous fixed-slice preference. This trend is opposite to the observed final outcome, where stronger SGD noise often leads to a higher probability of ending in flatter minima.

The resolution is that the final outcome is not a steady-state probability evaluated at an arbitrary fixed y. The landscape evolves while escape rates collapse. The probability that matters is the quasi-steady bias at the last point where inter-valley hopping remains effective. Stronger noise can therefore reduce the fixed-slice selectivity and still enhance final flat-minimum selection, because it moves the freezing point to a later part of the landscape.

This distinction is the core of the transient-freezing mechanism. Static bias determines the direction of selection on a given landscape slice. Freezing determines which slice is inherited as the final basin choice. Both ingredients are needed to explain why noise promotes flat minima during real training.

III. QUANTITATIVE PREDICTIONS AND EXPERIMENTAL TESTS

The theory becomes predictive when the final occupation is written as a stopping-time quantity. Let y_f be the training coordinate at which inter-valley hopping becomes negligible. Then

$$P^{\rm tr}_{\rm flat} \approx P^{\rm ss}_{\rm flat}(y_f)\,. \tag{9}$$

This expression does not assume that training reaches a global steady state. It assumes only that, before freezing, the transverse valley coordinate relaxes faster than the downhill coordinate, so the last quasi-steady bias is inherited by the final basin.

The freezing coordinate is determined by a competition of time scales. The inter-valley escape time is $\tau_{\rm esc} \sim (k_{f\to s} + k_{s\to f})^{-1}$, whereas the landscape changes on a drift time $\tau_y \sim |\dot{y}|^{-1}$. Freezing occurs when $\tau_{\rm esc}$ becomes longer than the time over which the quasi-steady distribution changes appreciably. In the two-valley model, this criterion gives, up to slowly varying prefactors,

$$y_f \approx y_b \frac{\sqrt{\Delta_S \ln\left(\Delta_S/\varepsilon^2\Phi^2\right)}}{x_2 - \sqrt{\Delta_S \ln\left(\Delta_S/\varepsilon^2\Phi^2\right)}}, \tag{10}$$

where ε defines the arrest criterion and Φ collects model parameters that vary slowly compared with the escape rates. The important consequence is monotonic: over the relevant range, larger Δ_S pushes the freezing point to larger y.

Substituting this freezing point into Eq. (9) yields a final flat-valley probability that increases with Δ_S for $\gamma > 1$,

$$P_{\rm flat}^{\rm tr} \approx \left[1 + \gamma^{-\frac{1}{2}} \left(\frac{\sqrt{\Delta_S}}{\varepsilon\Phi}\right)^{1-\gamma}\right]^{-1}. \tag{11}$$

Thus, the model predicts the empirical trend that stronger noise promotes flat-minimum selection, even though the fixed-slice selectivity alone would suggest the opposite.

The framework makes three direct predictions. First, increasing SGD noise should delay freezing over the stable training range. Second, later freezing should enlarge the accessible set of final solutions and shift that set toward flatter endpoints. Third, different optimizers or schedules should become comparable when they generate the same ordering of time scales: early inter-valley relaxation followed by barrier-driven arrest. These predictions can be tested by comparing freezing time, final-solution spread, flatness, and flat-valley probability across learning rates and batch sizes.

IV. DISCUSSION

We have shown that flat-minimum selection in SGD can be viewed as a nonequilibrium freezing problem. The central mechanism is that SGD first remains mobile among competing valleys and then becomes committed when inter-valley motion stops. The final basin is therefore selected at a finite stopping time, not only by a late-time steady state inside an already chosen valley.

This perspective clarifies the role of noise. SGD noise is not merely a scalar regularizer that always increases an instantaneous preference for flatness. Its main role is dynamical: it controls the duration of the exploratory window before basin commitment. Stronger noise can keep the trajectory mobile until later in training, where the evolving landscape and the geometry-induced bias together make selection of the flatter valley more likely.

The mechanism also gives a design principle. Learning-rate and batch-size schedules should be understood as tools for controlling the timing of basin commitment, not only as tools for guaranteeing convergence. A schedule that preserves inter-valley mobility too briefly may freeze into a sharp basin; a schedule that preserves it too long may prevent reliable convergence. The useful objective is to regulate the freezing time so that exploration persists until favorable valleys are accessible and then arrests before late-time noise destroys refinement.

The model is deliberately minimal, so its scope conditions are explicit. The transient-freezing picture should apply when there are multiple competing valleys, geometry-coupled stochasticity, barriers that evolve during descent, and a finite time window in which valley exchange is possible. These ingredients are not specific to the small network used here and are expected in richer architectures and datasets. What may change across systems is the most informative continuation protocol, endpoint-similarity measure, and dimensionality of the effective valley-coordinate space.

More broadly, the results connect the implicit bias of SGD to a class of nonequilibrium selection problems in which the outcome depends on the pathway to arrest. Fixed-landscape effective potentials remain useful because they determine the local direction of the bias. They become predictive of the final solution only when combined with the freezing time at which that bias is locked in. From this viewpoint, SGD selects flat minima not simply because they are statically favored, but because noise delays the freezing point at which basin competition is resolved.

DATA AND CODE AVAILABILITY

The code used for the empirical experiments and the two-valley simulations is available at `https://github.com/YangNing1995/Transient_Dynamics_SGD`.

ACKNOWLEDGMENTS

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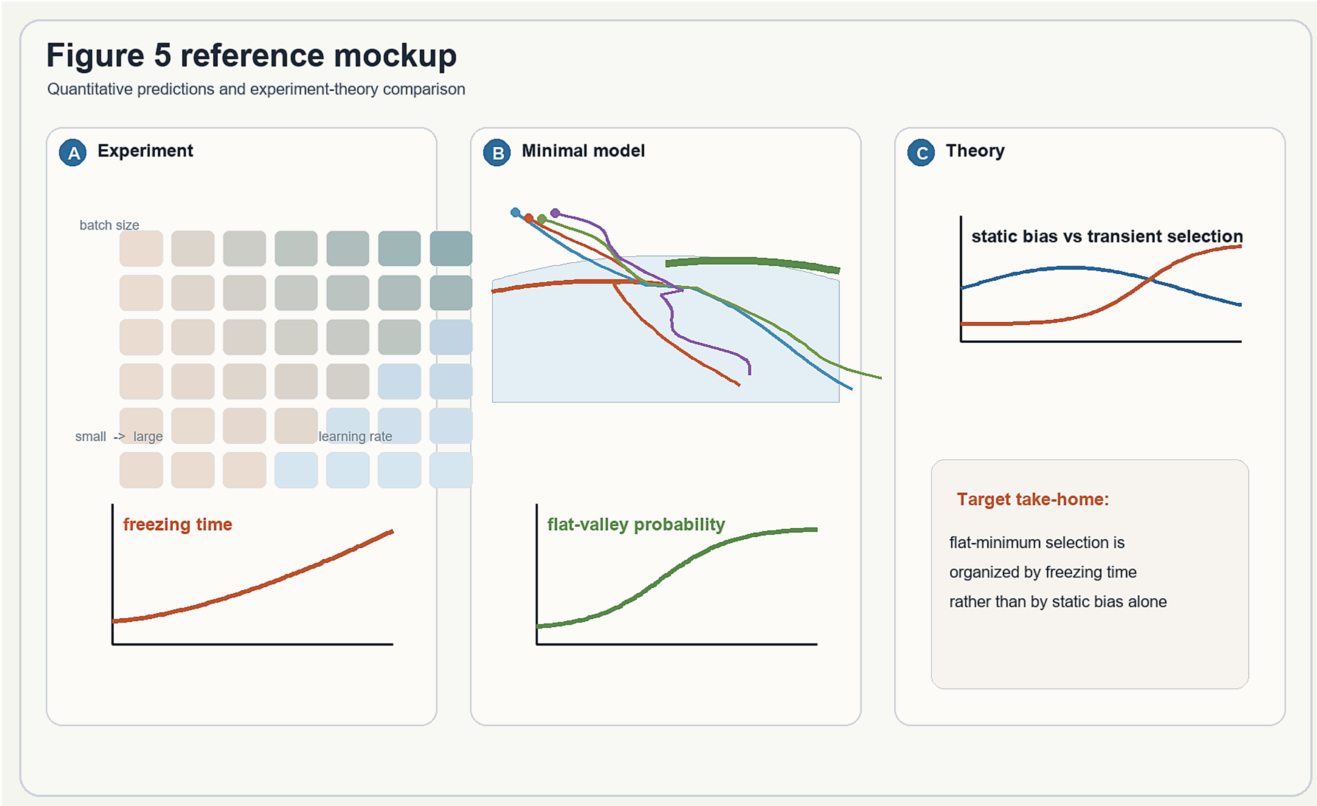

FIG. 5. Predictions and experiment-theory comparison. (a) The transient theory predicts that stronger effective noise delays freezing. (b) Evaluating the quasi-steady flat-valley bias at the freezing point predicts an increasing final flat-valley probability with noise. (c) Minimal-model simulations test the predicted relation between Δ_S, t_f, and P_{flat}. (d) Neural-network data are organized by the same variables: final-solution spread, final flatness, and upper-envelope test performance are compared with the measured freezing time. This comparison tests whether t_f is the stopping-time variable that organizes the inherited final-solution ensemble.

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